Boyle-Mariotte's law: how to isolate each quantity

I want V_2

1. Write the law. $$p_1 \cdot V_1 = p_2 \cdot V_2$$
2. p_2 is **multiplying** V_2: divide **both sides** by p_2. $$\frac{p_1 \cdot V_1}{p_2} = \frac{\cancel{p_2} \cdot V_2}{\cancel{p_2}}$$
3. Result: $$V_2 = \frac{p_1 \cdot V_1}{p_2}$$

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I want V_1

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2. p_1 is **multiplying** V_1: divide **both sides** by p_1. $$\frac{\cancel{p_1} \cdot V_1}{\cancel{p_1}} = \frac{p_2 \cdot V_2}{p_1}$$
3. Result: $$V_1 = \frac{p_2 \cdot V_2}{p_1}$$

I want p_1

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2. V_1 is **multiplying** p_1: divide **both sides** by V_1. $$\frac{p_1 \cdot \cancel{V_1}}{\cancel{V_1}} = \frac{p_2 \cdot V_2}{V_1}$$
3. Result: $$p_1 = \frac{p_2 \cdot V_2}{V_1}$$

Shortcut: what is **multiplying** on one side goes to the other side **dividing**.